

# Lauren Ostrowski

Medical Scientist Training Program  
University of California, San Diego  
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🔍 [google scholar](#)

## EDUCATION

### University of California San Diego

*M.D./Ph.D., Neurosciences Graduate Program*

2020 – present  
San Diego, CA, USA

- **Supervisors:** Dr. Timothy Gentner and Dr. Vikash Gilja

### Brown University

*B.Sc., Computational Neuroscience with Honors*

2014 – 2018  
Providence, RI, USA

- Honors distinction for thesis work in developmental epilepsy

## RESEARCH POSITIONS

### Gentner and Translational Neuroengineering Labs

*PhD Student*

2022 – present  
UCSD, San Diego, CA, USA

- Studying multi-regional neural population dynamics underlying the biomechanical control of learned vocal motor sequences in the songbird model system for speech production
  - *Developed novel neurosurgical techniques, custom software, and custom 3D-printed hardware for Neuropixels 2.0 probes, obtaining large-scale recordings of simultaneous single-unit activity in songbird HVC and RA across hundreds of song renditions*
- **Research residency at the Neural Interfacing Lab (PI: Christian Herff), Maastricht University, The Netherlands (March – July 2025, funded by the Merkin Graduate Fellowship)** investigating the neural mechanisms of bilingual speech production and low-dimensional dynamical regimes of language selectivity in intracranial recordings in Dutch-English speakers
  - *Designed two complementary paradigms, picture naming and unscripted dialogue with synthetic interlocutors, to elicit bilingual speech under ecologically valid conditions*
- Investigating temporal dynamics of linguistic and paralinguistic feature processing during speech perception across native and non-native languages using magnetoencephalography (MEG) and time-resolved neural decoding

### Halgren Lab

*Rotation Student*

2021  
UCSD, San Diego, CA, USA

- Explored neural activity in the human brain during speech perception using intraoperative data from high-density micro-ECoG electrodes placed over the superior temporal gyrus
- Using custom MATLAB-based analysis, found that phoneme-specific high-frequency oscillations phase-locked and co-occurred when phonemes were bound in consonant-vowel pairs

### Computational Neural DNA Dynamics Lab

*Rotation Student*

2020  
UCSD, San Diego, CA, USA

- Studied the frequency of atypical allele counts in healthy brain tissue using cytosine DNA methylation and chromatin profiles of post-mortem human frontal cortex tissue
- Developed and validated a pipeline in Python to detect partial aneuploidy at the single cell level

### Angelman Research Team at Biogen

*Data Analyst*

2019 – 2020  
Cambridge, MA, USA

- Analysis of EEG data in Angelman Syndrome and design of a robust, non-invasive EEG biomarker for Phase 1 clinical trial
- EEG findings eliminated the need for serial lumbar punctures in deployment of clinical trial

## BrainGate Consortium

2018 – 2019

### Research Assistant

Brown University and VA Medical Center, Providence, RI, USA

- Engineered software for neural signal processing with multi-institutional distribution
- Analyzed EEG activity in patients with recent stroke for future implementation in brain-computer interface

## Chu Lab

2016 – 2018

### Research Assistant

Massachusetts General Hospital, Boston, MA, USA

- Combined non-invasive physiological imaging techniques and behavioral data to identify biomarkers associated with seizure activity and neuropsychological impairments in developmental epilepsy
- Found correlation between white matter microstructure in language regions and comorbid language impairments
- Modeled whole-brain structural connectivity and analyzed network responses to electrical stimulation

## The Feinstein Institute for Medical Research

2015 – 2016

### Research Assistant

New Hyde Park, NY, USA

- Analyzed efficacy and interventional design of long-term health program to combat obesity and type 2 diabetes

## GRANTS AND AWARDS

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|      |                                                                               |
|------|-------------------------------------------------------------------------------|
| 2026 | UCSD Neurosciences Graduate Access and Enrichment Grant                       |
| 2024 | Sanford Institute for Empathy and Compassion Research Fellowship              |
| 2024 | UCSD Diversity in Neuroscience Enrichment Grant                               |
| 2023 | Merkin Graduate Fellowship                                                    |
|      | <i>Supported research residency at Maastricht University, The Netherlands</i> |
| 2023 | ACM SIGHPC Computational & Data Science Fellowship                            |
| 2018 | Sigma Xi Scientific Research Honor Society                                    |
| 2018 | Bachelor of Science in Neuroscience with Honors Designation                   |
| 2017 | Presidential LINK Award, Brown University                                     |
| 2014 | U.S. Presidential Scholar semi-finalist                                       |
| 2014 | New Hampshire Red Sox Service Scholarship                                     |
| 2014 | Milford Rotary Scholarship                                                    |
| 2014 | Joshua Sayvon Scholarship                                                     |

## PUBLICATIONS

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- [1] Weger P, Gimble S, Ottenhoff M, Verwoert M, **Ostrowski L**, ..., Kubben P, Herff C. Insular motor and error encoding supports self-correcting intracranial brain-computer interface. *Nat Commun* 2026; in press.
- [2] **Ostrowski LM**, Méndez JM, Tostado-Marcos P, Cooper BG, Gentner TQ. Automated inference of respiratory and syringeal biomechanical trajectories from birdsong acoustics. *J Neurophysiol* 2026; in press.
- [3] Tostado-Marcos P, Arneodo EM, **Ostrowski LM**, Brown DE II, ..., Gentner TQ, Gilja V. Population dynamics in songbird RA and HVC during learned motor-vocal behavior. *J Neurosci* 2026;e0580252026.
- [4] Hardstone R, **Ostrowski LM**, Dusang AN, López-Larraz E, ..., Lin DJ. Extension of voxel-based lesion mapping to multidimensional neurophysiological data. *Sci Rep* 2025;15:41488.
- [5] **Ostrowski LM**, Chinappen DM, Stoyell SM, Song DY, Ross EE, ..., Chu CJ. Children with Rolandic epilepsy have micro- and macrostructural abnormalities in white matter constituting networks necessary for language function. *Epilepsy Behav* 2023;144:109254.
- [6] Chinappen DM, **Ostrowski LM**, Spencer ER, Kwon H, Kramer MA, ..., Chu CJ. Decreased thalamocortical connectivity in resolved Rolandic epilepsy. *Clin Neurophysiol* 2023;153:21-27.
- [7] Spencer ER, Shi W, Komorowski RW, Gilbert JP, **Ostrowski LM**, ..., Chu CJ. Longitudinal EEG model detects antisense oligonucleotide treatment effect and increased UBE3A in Angelman syndrome. *Brain Commun* 2022;4(3):fcac106.
- [8] **Ostrowski LM**, Spencer ER, Bird LM, Thibert R, Komorowski RW, Kramer MA, Chu CJ. Delta power robustly predicts cognitive function in Angelman syndrome. *Ann Clin Transl Neurol* 2021;8(7):1433-1445.

- [9] Crocker B, **Ostrowski LM**, Williams ZM, Dougherty DD, Eskandar EN, . . . , Cash SS, Paulk AC. Local and distant responses to single pulse electrical stimulation reflect different forms of connectivity. *Neuroimage* 2021;237:118094.
- [10] Kramer MA, Stoyell SM, Chinappen DM, **Ostrowski LM**, Spencer ER, . . . , Chu CJ. Focal sleep spindle deficits reveal focal thalamocortical dysfunction and predict cognitive deficits in sleep activated developmental epilepsy. *J Neurosci* 2021;41(8):1816-1829.
- [11] Stoyell SM, Baxter BS, McLaren J, Kwon H, Chinappen DM, **Ostrowski LM**, . . . , Chu CJ. Diazepam induced sleep spindle increase correlates with cognitive recovery in a child with epileptic encephalopathy. *BMC Neurol* 2021;21(1):355.
- [12] Thorn EL, **Ostrowski LM**, Chinappen DM, Jing J, Westover MB, . . . , Chu CJ. Persistent abnormalities in Rolandic thalamocortical white matter circuits in childhood epilepsy with centrotemporal spikes. *Epilepsia* 2020;61(11):2500-2508.
- [13] **Ostrowski LM**, Song DY, Thorn EL, Ross EE, Stoyell SM, . . . , Stufflebeam SM, Chu CJ. Dysmature superficial white matter microstructure in developmental focal epilepsy. *Brain Commun* 2019;1(1):fcz002.
- [14] **Ostrowski LM**, Speiser PW, Accacha S, Altshuler L, Fennoy I, . . . , Rosenbaum M. Demographics and anthropometrics impact benefits of health intervention: Data from the Reduce Obesity and Diabetes project. *Obes Sci Pract* 2019;5(1):46-58.
- [15] Kramer MA, **Ostrowski LM**, Song DY, Thorn EL, Stoyell SM, . . . , Chu CJ. Scalp recorded spike ripples predict seizure risk in childhood epilepsy better than spikes. *Brain* 2019;142(5):1296-1309.
- [16] Song DY, Stoyell SM, Ross EE, **Ostrowski LM**, Thorn EL, . . . , Chu CJ. Beta oscillations in the sensorimotor cortex correlate with disease and remission in benign epilepsy with centrotemporal spikes. *Brain Behav* 2019;9(3):e01237.

## TEACHING

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|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>2023 – present</b> | Founder and Workshop Leader, Brain Machine Interface AI/ML Hackathon, <i>UCSD</i>                                                 |
| <b>2023 – 2025</b>    | Discussion Leader, Neurosciences Graduate Program Journal Club, <i>UCSD</i>                                                       |
| <b>2024</b>           | Teaching Assistant, Minor Proposition, <i>UCSD</i>                                                                                |
| <b>2022</b>           | Guest Lecturer, <i>UCSD School of Medicine</i><br><i>Lecture: "Health Care for People Experiencing Homelessness in San Diego"</i> |
| <b>2016</b>           | Teaching Assistant, General Chemistry, <i>Brown University</i>                                                                    |

## MENTORSHIP

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| <b>2022 – present</b> | Undergraduate Research Mentoring |
| <b>2020 – present</b> | UCSD Pre-Med Mentoring Program   |

## SERVICE

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| <b>2025 – present</b> | Teacher and Curriculum Developer: Las Colinas Detention Center<br><i>Prepare and teach classes at grant-funded science education program for incarcerated individuals in Santee, CA; in English and Spanish</i> |
| <b>2022 – 2025</b>    | Neurosciences Graduate Program Committees: Chair, NGP Student Wellness Committee; Member, NGP Diversity Committee                                                                                               |
| <b>2021 – 2022</b>    | Chapter President: Association of Women Surgeons, UCSD Student Chapter                                                                                                                                          |
| <b>2021 – 2022</b>    | Leadership & Clinical Volunteer: Homeless Health Initiative & Safe Haven Clinic                                                                                                                                 |
| <b>2021 – 2022</b>    | Public Health Advisory Council, Climate Action Campaign<br><i>Social Media Manager, patient health advocate at SD County Board of Supervisors meetings</i>                                                      |
| <b>2021 – 2022</b>    | Operation Harm Reduction<br><i>Volunteer distributing Narcan, clean needles, and first aid supplies</i>                                                                                                         |
| <b>2020 – present</b> | Medical Scientist Training Program Committees: Member, MSTP Diversity Committee; Member, MSTP LGBTQ+ Health Committee                                                                                           |
| <b>2020 – 2022</b>    | Executive Committee: LGBTQ+ Pharmacy & Medicine<br><i>Organized SoCal LGBTQ+ Conference, Trans Week of Visibility, UCSD Rainbow Pin Ceremony</i>                                                                |

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| <b>2020 – 2022</b> | Executive Committee: Border Health Project<br><i>Organized mobile health clinics, supply drives for migrant farmworkers</i>           |
| <b>2020 – 2022</b> | Anti-Racism Coalition at UCSD<br><i>Policy &amp; Admissions Subcommittees: combating ICE in healthcare, microaggression reporting</i> |
| <b>2020 – 2021</b> | Clinic Manager: UCSD Women's Free Clinic<br><i>Managed clinic operations and planned contraceptive counseling services</i>            |
| <b>2017 – 2019</b> | Clinical Volunteer: Clínica Esperanza / Hope Clinic, Providence, RI                                                                   |
| <b>2017 – 2018</b> | President: Women in Science and Engineering at Brown                                                                                  |
| <b>2017 – 2018</b> | Mentor: The Adolescent Leadership Council, Providence, RI<br><i>Adolescent chronic illness support group</i>                          |
| <b>2015 – 2018</b> | Peer Advisor: The Meiklejohn Peer Advising Program, Brown University                                                                  |

## PRESENTATIONS

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- [1] Society for the Neurobiology of Language 2026, Geneva, Switzerland. (September 29 – October 2, 2026). Poster: "Asymmetric dynamical regimes underlie bilingual language control in intracranial recordings from Dutch-English speakers." **L. Ostrowski**, et al.
- [2] Computational and Systems Neuroscience (COSYNE) 2026, Lisbon, Portugal. (March 12 – 17, 2026). Poster: "Transient, behavior-dependent coupling of neural population dynamics during learned motor sequences." J. Gorman\*, **L. Ostrowski\***, et al. [\*equal contribution]
- [3] Society for Neuroscience 2025, Chicago, IL, USA. (November 15 – 18, 2025). Poster: "Cortical temporal dynamics of speech perception: Dissociating linguistic and paralinguistic elements." J. Xing, **L. Ostrowski**, et al.
- [4] Society for Neuroscience 2025, Chicago, IL, USA. (November 15 – 18, 2025). Poster: "Latent premotor neural dynamics reveal species-specific vocal strategies in songbirds." J. Gorman, **L. Ostrowski**, et al.
- [5] Society for Neuroscience 2025, Chicago, IL, USA. (November 15 – 18, 2025). Poster: "Species differences in real-time sensorimotor error processing during vocalization." L. Stanwicks, **L. Ostrowski**, et al.
- [6] American Society of Neuroradiology 2023, Los Angeles, CA, USA. (March 15, 2023). Poster: "Linking post-stroke neurophysiology to neuroanatomy: Novel method to extend voxel-lesion mapping to multi-dimensional EEG data." R. Hardstone, **L. Ostrowski**, et al.
- [7] American Society of Neuroradiology 2022, Boston, MA, USA. (April 1, 2022). Poster: "Persistent asymmetry of aperiodic resting-state neural activity in cortical and subcortical strokes." R. Hardstone, **L. Ostrowski**, et al.
- [8] Society for Neuroscience 2018, San Diego, CA, USA. (November 3 – 7, 2018). Poster: "Focal abnormalities in white matter microstructure correspond to neuropsychological comorbidities in BECTS." **L. Ostrowski**, et al.
- [9] American Epilepsy Society 2017, Washington DC, USA. (December 1 – 5, 2017). Poster: "Dysmature white matter microstructure in developmental epilepsy." **L. Ostrowski**, et al.
- [10] Endocrine Society 2016, Boston, MA, USA. (April 1 – 4, 2016); Obesity Society 2017, Washington, DC, USA. (October 29 – November 2, 2017). Poster: "Demographics and anthropometrics impact benefits of health intervention: Data from the ROAD Project." **L. Ostrowski**, et al.

## SKILLS

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**Programming languages:** Python, MATLAB, C++

**Scientific computing & ML:** PyTorch, TensorFlow/Keras, NumPy, SciPy, Scikit-learn, statsmodels, Pandas, generalized linear models, dimensionality reduction, neural decoding, dynamical-systems analysis

**Neurophysiology & signal processing:** spike sorting (Kilosort), SpikeGLX/Open Ephys, NWB, MNE-Python, spectral/signal processing, DTI/tractography (FSL, MRtrix, FreeSurfer, Probtrackx)

**Hardware & fabrication:** 3D printing, Solidworks, Onshape

**Tools:** Git, Linux/Unix systems, HPC/cluster computing (SLURM), Containers (Docker), AWS, Adobe Illustrator, Adobe Photoshop, LaTeX

**Languages:** English, Spanish